



ShaniOS: Making Linux Updates as Safe as Flipping a Light Switch

This new immutable Linux distribution eliminates system update anxiety and promises to make Linux desktop maintenance stress-free.

Picture this: You're working on an important project when a system update notification appears. Do you install it immediately and risk breaking something, or postpone it indefinitely? For most Linux users, system updates remain a persistent source of anxiety. ShaniOS, a new Arch-based distribution, aims to transform this experience by making updates as safe and predictable as flipping a light switch.

The problem with traditional updates

Traditional Linux distributions rely on package managers like APT, DNF, or Pacman that modify the running system in place. While this approach works most of the time, it occasionally leaves systems in broken states—particularly when power outages occur during updates or conflicting packages are installed.

"I've witnessed too many systems break during updates, causing hours of downtime for users," explains Shrinivas Vishnu Kumbhar, creator of ShaniOS and co-founder of Garuda Linux. "The solution isn't to avoid updates—it's to make them bulletproof."

Enter blue-green deployment for desktops

ShaniOS adopts a battle-tested DevOps technique called 'blue-green deployment' for desktop use. Rather than modifying your current system, it maintains two identical copies of your operating system on separate Btrfs subvolumes labelled *@blue* and *@green*. During updates, changes are applied to the inactive copy, tested, and then you simply switch to the updated version on your next reboot.

If something goes wrong, you just switch back—the entire process takes less than a minute.

How it works

In ShaniOS, system updates are managed through a single command:

```
sudo shani-deploy
```

This command:

- Downloads updates to the inactive system copy.
- Applies them safely without affecting your running system.
- Configures the bootloader to use the updated copy.
- Provides instant rollback capability if the update doesn't work properly.

Your applications, documents, and personal settings remain completely untouched throughout the process.

Applications: Isolated but integrated

Since the core system is immutable (read-only), applications require a different approach. ShaniOS employs